PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systematic review and meta-analysis
AUTHORS	Foley, Louise; Larkin, James; Lombard-Vance, Richard; Murphy, Andrew; Hynes, Lisa; Galvin, Emer; Molloy, Gerard

VERSION 1 – REVIEW

REVIEWER	Fowler Davis, Sally Sheffield Hallam University, Faculty of health and Wellbeing
REVIEW RETURNED	10-Nov-2020

GENERAL COMMENTS	Thanks for inviting me to review, I found the article well written and methodogically robust. The presentation was problematic however and the long lists of numbers could be revised perhaps into a table to make the list of criteria more readable. You have reported on the increased burden of morbidity and medication but not on the effect of deprescribing which is a current and important issue for health services where medicines optomisation is a priority- see https://doi.org/10.1093/ageing/afy206.02. The results are interesting although the focus on frequency of studies is the dominant means of analysis and in the discussion, it would be important to focus on the significance of non-adherance as a critical choice for some patients and the notion of complex case management is perhaps the missing element in the conclusion. I am not sure whether the metaphor 'uneven landscape' is useful and perhaps a clearer description would be helpful, or another idiom that is perhaps better known- 'a mixed bag!' The focus on and call for consitent measurement of non-adherance as a primary measure of outcome is welcom and I agree.

REVIEWER	Calderón-Larrañaga, Amaia Karolinska Institutet
REVIEW RETURNED	15-Nov-2020

GENERAL COMMENTS	The paper aims to systematically review the prevalence and predictors of medication non-adherence in multimorbid subjects. The research question is pertinent, data are creatively analysed, and the paper is nicely written. Despite the heterogenous landscape pictured by the review (which was expectable given the broad inclusion criteria), the authors still provide some interesting insight for future research, such as the need to assess multiple medication adherence in single persons, given that medication adherence importantly varies across conditions and medications. I have only two comments that the authors may address. First, I did not see any information in terms of the setting where studies were

	carried out. The prevalence of non-adherence can be different in community settings versus clinical or nursing home settings, but this analytical axis is lacking in the study. In the same way systematic reviews on the prevalence of multimorbity have differentiated between study settings, I invite the authors to follow the same rule in their review. Second, incidentally, a paper I was involved in was not included in the review and I was not able to identify the reasons for its exclusion (https://pubmed.ncbi.nlm.nih.gov/26814635/), which made me think the inclusion/exclusion criteria may need to be more accurately defined. The authors mention that “the presence of multimorbidity was required to be part of study aims and/or inclusion criteria”; while it was not an inclusion criterion of our study, if was its main aim.

REVIEWER	Dregan, Alexandru King's College London, UK, Primary Care and Public Health Sciences
REVIEW RETURNED	13-Mar-2021

GENERAL COMMENTS	This is a worthy investigation that highlights concerns around prescribing in patients with multimorbidity. Major strength include the use of large number of studies, though these can also be a limitation since it dilutes the interpretation of the evidence. In this Reviewer opinion, a more focused review based on specific criteria (ie self-reports, EHRs, trials) would be of greater practical relevance and offer clear guidelines for future studies. Equally, age of patients was not considered and one cannot make sweeping comments across adults with diverse degree of impaired cognition, social isolation and frailty. Additionally, it's not clear whether authors used polypharmacy as a definition of multimorbidity to identify relevant studies. Introduction Page 1, Line 38 - if the patient has a single disease only - how is the artificial underestimation statement relevant? Need to place this sentence within the context of multimorbidity Methods In this Reviewer opinion, doctoral theses might not necessarily have been published and I would be reluctant to include as evidence Page 6, Line 6 - Not clear what the authors imply by ten year range as a definition of multimorbidity? Why should year range define multimorbidity? There are papers published in the same year, yet using different conceptualisations of the term Please clarify whether the review included both observational and clinical trials evidence -indeed why not focus on the latter? Page 6, Line 48: - Des that mean that the review refers only to observational studies and clinical trials were excluded from the review? Page 7, Lines 12-14: The decision to exclude studies looking at combination of diseases only needs more detail - are you referring to comorbidity? If both diseases are equally important, they should nevertheless be considered multimorbid, unless a cut-off point of 3 conditions is used.

	Did the review include studies that referred to both mental and physical multimorbidity, or just physical multimorbidity? Was the age range of study participants considered for eligibility criteria? Results Page 8, Lines 11-15: These sentences are poorly structured and confusing - apart from the different choice of words, they all imply the same number of conditions. Why complicate things and not just simply say, 2 or more conditions? First Paragraph - authors should be more selective in referencing, provide exemplars and clarify in the table which category each study belongs to. No need to list all studies in the text. Page 9, last paragraph: This Reviewer found the description of non-adherence by number of conditions confusing and unclear. Didn't the authors use multimorbidity as a definition? Why the need to focus on non-adherence by single condition? Are you going to compare non-adherence between studies with and without multimorbidity for a specific condition? Page 10, Physical and mental comorbidities (should be multimorbidity?): Why were there differences in the prevalence of non-adherence between studies? Without expanding on these differences, you are not really telling the reader much. The Results section should have been structured around the issues of age of participants, number of co-existing diseases, nature of co-existing diseases, method of assessment etc.. As it currently stands these important details are lost. Discussion Page Page 14, Line 43: In the Results section it was stated that there were no differences by mental and physical conditions? Page 20, Lines:28-32: I don't follow the logic of the sentence related to depression/anxiety - why would self-report exclude a diagnosis of depression/anxiety? Might be the opposite, as people with depression/anxiety are poorly captured in medical records.

REVIEWER	Zhou, Lili AbbVie Inc, Health economics and outcome research
REVIEW RETURNED	11-Apr-2021

GENERAL COMMENTS	This manuscript entitled “Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systemic review and meta-analysis” added to the literature in examining the medication non-adherence among individuals with multimorbidity, regardless of medications and conditions to be investigated. The strengths of this work include identifying an inclusive list of eligible studies. While I appreciate the efforts that investigators devoted to this work, quite a few major issues substantially dampen my enthusiasm for this study. As investigators acknowledged in the study, medication non-adherence varied significantly by medications and conditions to be investigated. Furthermore, medication non-adherence can be impacted by many other factors like the reimbursement policies distinct in countries/regions.

	Ignoring the substantial heterogeneity of the included studies and providing a pooled overall non-adherence prevalence is not informative from my perspective. Investigating a more granular level of medication non-adherence based on the patients, conditions, and medications is more of interest in my point. There are several notable major concerns in the manuscript needed to be clarified in order to ensure the validity of the findings. Areas could be improved to strengthen the manuscript and study are listed below. 1. (Page 3, abstract, methods): Investigators stated the databases that were searched for relevant studies. Why did not the investigators search the Cochrane Library (https://www.cochranelibrary.com/cdsr/about-cdsr) ? It is an important data source to check whether any similar systemic review and meta-analysis has been conducted previously. If so, investigators should mention that in the introduction to clarify what incremental values that this manuscript will provide. There may be some systemic review studies that have investigated the similar research question, but I did not see investigators mentioning them in the introduction. Failing to do that weakens the importance of this study. Furthermore, the investigators may identify extra eligible studies through reviewing the Cochrane reviews. 2. (Page 3, abstract, methods): The time span of this review was between 2009 and April 2019 as indicated in the abstract. I understand that the purpose was to align with the protocol published in 2019. However, given the date of reviewing this study (April 2021), I strongly suggest investigators performing an updated search looking till the current date. 3. (Page 5, introduction): Investigators did not mention any of previous systemic review or meta-analysis studies that examined the medication non-adherence among people with multimorbidity. It is unclear what extra values that this study will provide if there is such type of study. Suggest investigators adding some contents to clarify that. Thanks! 4. (Page 6-7, data extraction and quality assessment): Investigators discussed that they would assess the quality of each study, but no studies would be excluded based on the quality. An overall view of study quality was presented in Figure 2. However, I am not sure of the value of assessing the study quality if the investigators did not exclude those studies with low-quality or did not do any subgroup analysis in reporting the results in high- and low-quality studies separately. From my perspective, reporting and synthesizing results from low-quality studies is pointless due to their untrustworthy findings. I understand that investigators would like to include an inclusive list, but a balance between completeness and study quality needs to be achieved. If investigators want to include all the studies, I suggest stratifying the studies based on quality and then reporting the results separately. In addition, Figure 2 just provided an overall view of study quality. Suggest adding one more column in the study characteristics table in the supplement to indicate the quality of each study. Last, it is difficult to read the assessment items in Figure 2. Please make the font size bigger.

	5. (Page 7, data analysis): It claimed that “Meta-analysis was conducted with studies which recruited participants as ‘people living with >= 2 chronic conditions’ and not studies which recruited participants with combinations of specific conditions”. I agree that a quantitative synthesis including all studies was not appropriate given the substantial heterogeneity between studies. However, I am not sure why meta-analysis was conducted only for these studies as well as the value of results. The investigators claimed that these studies reflected the most inclusive definition of multimorbidity, but based on my observations, the medication non-adherence varies substantially by medications and conditions to be investigated. Many other factors will also affect the medication non-adherence (e.g., reimbursement policy in the corresponding country/region, patient demographics). Providing an overall pooled prevalence of medication non-adherence regardless of specific medications and indications is not informative from my perspective. I cannot see the value of these findings. Investigators could look into a more granular level of medication non-adherence in different types of patients, conditions, and medications. I think those results are more of interest compared to the present results in the manuscript. Suggest investigators conducting more subgroup analyses. 6. (Page 8, study characteristics; Page 9, prevalence of medication non-adherence): It stated in “Study characteristics” that 19 studies recruited participants using an inclusive definition of multimorbidity, but under “Prevalence of medication non-adherence” only 8 studies were included in the meta-analysis. Could investigators clarify why 11 studies were excluded from the meta-analysis?

REVIEWER	Belli, Hayley NYU Langone Health, Population Health
REVIEW RETURNED	16-Apr-2021

GENERAL COMMENTS	Overall, a very thorough review and well written manuscript. The inclusion of search criteria and individual study data (even R code) demonstrate both meticulousness and transparency on behalf of the authors. My minor comments are the following: Figure 3 – results from the random effects model seem to just point to the largest study dominated by 10x the population of the other studies, but the non-adherence range among all 8 studies is considerable (<20% to 60%). Please walk the readers through the interpretation of this result give such high variability for a small number of studies. In the section, “Medication non-adherence over time” please summarize the endpoints for non-adherence across studies (e.g., what fraction of studies measured adherence in a few months, 6 months, 1 year, etc.), as it’s difficult to interpret increases and decreases without knowing the denominator. The discussion section raises the point that medication frequency or number of tablets of course impacts medication adherence. It would be helpful to summarize in the results any findings regarding differences in adherence among conditions that require higher medication frequency or tablet count, compared to conditions with lower medication frequency/table counts.

VERSION 1 – AUTHOR RESPONSE

	Reviewer comment	Author response	Changes to manuscript
	Reviewer 1		
1.1	Thanks for inviting me to review, I found the article well written and methodologically robust. The presentation was problematic however and the long lists of numbers could be revised perhaps into a table to make the list of criteria more readable.	Many thanks to this reviewer for your helpful comments, which are addressed below and in the revised manuscript. We have revised the manuscript to remove the long lists of references, and instead used examples of studies in each section as relevant. This is in line with similar advice from Reviewer#3 in comment 3.10 below. A section has been added to the supplementary material (file 6) outlining the full list of articles relevant to each section of the results. An example from one section of the results is provided to the right; please see manuscript for all related changes.	One-hundred and thirty-three studies included participants with ≥ 2 specific conditions, e.g. Gibson et al. recruited participants self-managing type 2 diabetes and hypertension;[1] 26 studies included participants with one specific condition and ≥ 1 other condition(s), e.g. Puyat et al. recruited participants living with depression and an additional chronic health condition;[2] and 19 studies recruited participants using an inclusive definition of multimorbidity, e.g. Schüz and colleagues[3] recruited people living with ≥ 2 conditions from the Charlson Comorbidity Index[4] or Functional Comorbidity Index.[5]
1.2	You have reported on the increased burden of morbidity and medication but not on the effect of deprescribing which is a current and important issue for health services where medicines optimisation is a priority- see https://doi.org/10.1093/ageing/afy206.02.	Thank you for highlighting the relevance of our review findings to this research area. We have added a sentence to our discussion to highlight the unclear relationship between medication burden and non-adherence that is highlighted both in our results and in the recent review of deprescribing interventions cited by the reviewer.	A recent systematic review concluded that deprescribing interventions do not consistently improve adherence,[6] suggesting an equivocal relationship between medication burden and non-adherence.
1.3	The results are interesting although the focus on frequency of studies is the dominant means of analysis and in the discussion, it would be important to focus on the significance of non-adherence	We agree that non-adherence may represent a critical choice for some patients, and consider this most relevant to our discussion of findings related to illness beliefs and intentional non-adherence.	This tendency to be influenced by illness beliefs may position intentional non-adherence as a rational judgement for some people self-managing chronic conditions; how this transpires as selective non-

	as a critical choice for some patients and the notion of complex case management is perhaps the missing element in the conclusion.	We have updated the manuscript to reflect this. We agree that the context of complex case management is relevant to the conclusion of this review, and have updated the manuscript to reflect this. Thank you for these suggestions which have strengthened our discussion of results.	adherence among people self-managing multiple conditions should be explored in future research. [7] Management of multimorbidity demands integration across conditions and clinical guidelines, requiring practitioners and researchers to focus on individual people as opposed to individual diseases.[8] While the move towards this multi-disease focus is welcomed, attempts by researchers and clinicians to improve medication adherence should determine for which conditions individuals require most support.
1.4	I am not sure whether the metaphor 'uneven landscape' is useful and perhaps a clearer description would be helpful, or another idiom that is perhaps better known- 'a mixed bag!' The focus on and call for consistent measurement of non-adherence as a primary measure of outcome is welcome and I agree.	We agree and have changed the expression used in this section to be more clear to readers, replacing 'uneven landscape' with 'complexity'.	This review summarises the complexity of non-adherence in multimorbidity, as well as factors that should be considered for the development of interventions to improve adherence for 'the most common chronic condition'.
Reviewer 2			
2.1	The paper aims to systematically review the prevalence and predictors of medication non-adherence in multimorbid subjects. The research question is pertinent, data are creatively analysed, and the paper is nicely written. Despite the heterogeneous landscape pictured by the review (which was expectable given the broad inclusion criteria), the authors still provide some interesting insight for future research, such as the need to assess multiple medication adherence in single persons, given that medication adherence importantly varies across conditions and medications. I have only two	Many thanks to this reviewer for your helpful comments, which are addressed below and in the revised manuscript.	n/a

	comments that the authors may address.		
2.2	First, I did not see any information in terms of the setting where studies were carried out. The prevalence of non-adherence can be different in community settings versus clinical or nursing home settings, but this analytical axis is lacking in the study. In the same way systematic reviews on the prevalence of multimorbidity have differentiated between study settings, I invite the authors to follow the same rule in their review.	We have included a column in the study characteristics table outlining the setting of included studies, where reported. We have also added a section to the supplementary material highlighting which individual studies were performed in primary care, secondary care etc. As also noted below in 4.8, we recognise the potential to complete a range of very interesting subgroup analyses using the data arising from this systematic review, including using setting data to stratify analyses; however this is not considered possible within the constraints of a single journal article. For this reason, we will make our comprehensive database available to the research community for collaboration on future sub-analyses, as has previously been done in a Cochrane review of medication adherence interventions reporting a similarly large number of studies with a wide range of potentially influential variables.[9]	Please see updated supplementary files 5 and 6. ***Addition to data sharing statement:*** Our systematic review produced a large amount of information and the arising database is available for future collaboration on additional analyses. Please contact the corresponding author with any inquiries.
2.3	Second, incidentally, a paper I was involved in was not included in the review and I was not able to identify the reasons for its exclusion (https://pubmed.ncbi.nlm.nih.gov/26814635/), which made me think the inclusion/exclusion criteria may need to be more accurately	Thank you for alerting us to this paper which does in fact meet our inclusion criteria. We have revisited our records in Covidence and found this paper was mistakenly excluded at title & abstract screening when a portion of the records were not screened independently by two	n/a

	defined. The authors mention that "the presence of multimorbidity was required to be part of study aims and/or inclusion criteria"; while it was not an inclusion criterion of our study, if was its main aim.	reviewers. This reflects a limitation addressed in our discussion outlining the potential for relevant articles to be excluded. This paper has now been added to the review results and synthesised with the other included studies. We also checked a random sample of 100 studies that were excluded at this stage of the review. This revealed no omitted papers.	
Reviewer 3			
3.1	This is a worthy investigation that highlights concerns around prescribing in patients with multimorbidity. Major strength include the use of large number of studies, though these can also be a limitation since it dilutes the interpretation of the evidence. In this Reviewer opinion, a more focused review based on specific criteria (ie self-reports, EHRs, trials) would be of greater practical relevance and offer clear guidelines for future studies. Equally, age of patients was not considered and one cannot make sweeping comments across adults with diverse degree of impaired cognition, social isolation and frailty. Additionally, it's not clear whether authors used polypharmacy as a definition of multimorbidity to identify relevant studies.	Many thanks to this reviewer for your helpful comments, which are addressed below and in the revised manuscript. We have added a paragraph to the discussion section of the manuscript acknowledging the unclear role of age in non-adherence, as well as the potential for different variables to predict non-adherence at different stages of the lifespan. Polypharmacy was not used as a definition of multimorbidity; however use of two or more medications that could be inferred as treatment for two or more chronic conditions was inferred as multimorbidity. This meant studies of people taking two or more medications were considered, however only if these medications were not used for a single condition. This has been clarified in the manuscript.	We did not stratify our analysis of prevalence by age, limiting the interpretation of findings which may be confounded by age-related experiences. It has been noted that there is no consensus on age as a predictor of non-adherence,[10] and age was not found to be a significant predictor of non-adherence in this review. However, it may be plausible that variables are differentially associated with non-adherence at different life stages. For example, Domino and colleagues[11] reported lower levels of non-adherence among those with more conditions, with a mean age among participants of 43 years; Schüz et al. found higher non-adherence among participants with more conditions, with a mean age of 73 years.[12] Exploring how non-adherence is variably predicted by clinical and psychosocial influences across the lifespan could inform intervention development which is appropriately targeted according to life stage. Polypharmacy was not used as a definition of multimorbidity; however studies referring to use of two or more medications that could be inferred as treatment for two or more chronic conditions were considered.

3.2	Introduction Page 1, Line 38 - if the patient has a single disease only - how is the artificial underestimation statement relevant? Need to place this sentence within the context of multimorbidity	Thank you for highlighting the ambiguity in this statement. The manuscript has been updated to clarify this in the context of multimorbidity.	Such a focus may produce an artificial underestimation of the complexity of chronic disease self-management when, in reality, most people with a chronic condition have additional morbidity(s).[13] This may lead to policies and interventions which are not designed with 'the most common chronic condition'[8] in mind.
3.3	Methods In this Reviewer opinion, doctoral theses might not necessarily have been published and I would be reluctant to include as evidence	While our review did not define a grey literature search strategy, it was decided to maintain doctoral theses identified through database searches in the review. The intended purpose is to reflect the unpublished literature in our review and attempt to reduce publication bias, as has been recommended previously for systematic reviews.[14]	n/a
3.4	Page 6, Line 6 - Not clear what the authors imply by ten year range as a definition of multimorbidity? Why should year range define multimorbidity? There are papers published in the same year, yet using different conceptualisations of the term	We thank the reviewer for highlighting the ambiguity in this statement. This statement has been clarified in the manuscript to emphasise that the ten year range reflects a notable increase in multimorbidity research over this period of time, with reference to a 2017 bibliometric study of the multimorbidity literature.	A ten year range was chosen to reflect the marked increase in multimorbidity literature over the last decade.[15]
3.5	Please clarify whether the review included both observational and clinical trials evidence -indeed why not focus on the latter?	The review aimed to synthesise observational data only, which included baseline data from clinical trials where relevant. This has been clarified in the manuscript. Observational data was considered most suitable to address the aims of the systematic review, i.e. to determine prevalence and predictors of non-adherence. Therefore clinical trial evidence, beyond baseline observational data, was not considered appropriate to address these aims.	Studies were not excluded based on design, however only observational data was extracted for review. Therefore clinical trials that did not report baseline non-adherence were excluded.
3.6	Page 6, Line 48: - Does that mean that the review refers only	See point 3.5	See point 3.5

	to observational studies and clinical trials were excluded from the review?		
3.7	Page 7, Lines 12-14: The decision to exclude studies looking at combination of diseases only needs more detail - are you referring to comorbidity? If both diseases are equally important, they should nevertheless be considered multimorbid, unless a cut-off point of 3 conditions is used.	This decision was taken out of consideration for the heterogeneity that would be produced from including these studies of pre-specified conditions in a meta-analysis. As is addressed later in comment 4.8, we acknowledge the heterogeneity that is nevertheless produced by the 8 studies which were included in the meta-analysis. However, these 8 studies are considered to be more focussed in addressing a universal experience of multimorbidity – albeit a multifaceted and complex experience across individuals, offering results that can be extrapolated for future attempts to address non-adherence in this population. We recognise that all may not take this same perspective. The manuscript has been updated to reflect our reasoning here.	Meta-analysis was conducted with studies which recruited participants as 'people living with ≥ 2 chronic conditions'. This group of studies was considered to be most focussed in addressing a universal experience of multimorbidity – albeit a multifaceted and complex experience across individuals – offering results that can be used in future attempts to address non-adherence in this population. Studies which recruited participants with pre-specified conditions were therefore not included in the meta-analysis.
3.8	Did the review include studies that referred to both mental and physical multimorbidity, or just physical multimorbidity? Was the age range of study participants considered for eligibility criteria?	We thank the reviewer for flagging the ambiguity in our reporting here. The review included studies of people living with mental and physical multimorbidity. This has been clarified in the manuscript. Eligibility criteria required all study participants to be 18 years or older. The manuscript has been updated to state this.	Quantitative studies including adults aged 18 years or older, living with two or more chronic conditions (including physical and/or mental health conditions) reporting medication non-adherence and/or predictors of non-adherence in an extractable format were considered.
3.9	Results Page 8, Lines 11-15: These sentences are poorly structured and confusing - apart from the different choice of words, they all imply the same number of conditions. Why complicate things	We acknowledge that this information amounts to the same outcome for each study, i.e. 2 or more conditions. However, this detail was included to highlight the different iterations of '2 or more conditions' that were studied, which is considered of	One-hundred and thirty-three studies included participants with ≥ 2 specific conditions, e.g. Gibson et al. recruited participants self-managing type 2 diabetes and hypertension;[1] 26 studies included participants with one specific condition and ≥ 1 other condition(s), e.g. Puyat et al.

	and not just simply say, 2 or more conditions?	relevance later in the manuscript when discussing studies included in the meta-analysis and analysis of predictors. In line with related comments 1.1 and 3.10, we have removed the list of references and provided exemplars to clarify this and make this section more informative.	recruited participants living with depression and an additional chronic health condition;[2] and 19 studies recruited participants using an inclusive definition of multimorbidity, e.g. Schüz and colleagues[3] recruited people living with ≥ 2 conditions from the Charlson Comorbidity Index[4] or Functional Comorbidity Index.[5]
3.10	First Paragraph - authors should be more selective in referencing, provide exemplars and clarify in the table which category each study belongs to. No need to list all studies in the text.	See points 1.1 and 3.9	See points 1.1 and 3.9
3.11	Page 9, last paragraph: This Reviewer found the description of non-adherence by number of conditions confusing and unclear. Didn't the authors use multimorbidity as a definition? Why the need to focus on non-adherence by single condition? Are you going to compare non-adherence between studies with and without multimorbidity for a specific condition?	We recognise the apparent vagueness of reporting in this section of the original manuscript. The review includes only participants with multimorbidity and we did not compare non-adherence between studies of participants with and without multimorbidity. However, many studies reported a single adherence estimate that was an average across all conditions, or only reported adherence for one condition studied. In this section of the results we are focussing only on studies that reported separate estimates for each condition studied. The manuscript has been updated to clarify this distinction, with examples included.	Within individual studies, the extent of non-adherence for one condition did not always correspond with non-adherence for another condition among people taking medications for multimorbidity. Twenty-six studies reported non-adherence prevalence estimates separately for each condition, e.g. Foguet-Boreu reported that, among people living with both diabetes and hypertension, 39.8% were non-adherent to medication for diabetes, while 23.8% were non-adherent to medications for hypertension.[16]
3.12	Page 10, Physical and mental comorbidities (should be multimorbidity?): Why were there differences in the prevalence of non-adherence between studies? Without expanding on these differences, you are not really telling the reader much.	We have changed this subheading to refer to multimorbidity instead of comorbidity and thank the reviewer for this suggestion. Within this section of the results, the reason for differences between studies was not clear, as no strong pattern was observed in relation to method of adherence	Physical and mental multimorbidity

		measurement, conditions studied or setting of the research. Any statement of reasons for differences between studies in the results section are considered overly tentative given the number of factors that may have influenced the results. However, potential reasons for inconsistent findings are explored in our discussion section of the original manuscript, including the cyclical nature of some mental health conditions.	
3.13	The Results section should have been structured around the issues of age of participants, number of co-existing diseases, nature of co-existing diseases, method of assessment etc.. As it currently stands these important details are lost.	We agree there are many ways the reporting of these results could have been approached. We have addressed the point on age of participants in 3.1 above, and its inconsistent role as a predictor of non-adherence; the role of number of conditions was reported in the analysis of predictors; in relation to the nature of co-existing conditions, we have included a section on physical and mental multimorbidity; and our results includes a dedicated section on method of adherence assessment.	n/a
3.14	Discussion Page 14, Line 43: In the Results section it was stated that there were no differences by mental and physical conditions?	We have updated the results section to clarify that the differences between physical and mental health conditions did not follow a consistent pattern across studies, as opposed to there being no difference between non-adherence for physical conditions and non-adherence for mental health conditions.	Differences in non-adherence between physical and mental health conditions were not consistent across the literature.
3.15	Page 20, Lines:28-32: I don't follow the logic of the sentence related to depression/anxiety - why would self-report exclude a diagnosis of depression/anxiety? Might be the opposite, as people with depression/anxiety are poorly captured in medical records.	We included studies that used scores on a self-report measure as an indicator of depression/anxiety using measures such as Beck Depression Inventory,[17] Patient Health Questionnaire[18] etc. This might lead to some participants being identified as having depression/anxiety without having	Thirdly, studies of people with depression or anxiety were eligible for inclusion if the condition was identified through a self-report measure. As a result, it is possible that not all participants had a clinical diagnosis. This decision was made to account for the overrepresentation and underdiagnosis of depression among people with chronic

		a recorded clinical diagnosis. The aim here was to represent those living with depression/anxiety who are not captured in medical records. We have updated the section to clarify this.	conditions due to incomplete recording in medical records,[19] and has been made previously when reviewing the multimorbidity literature.[20]
Reviewer 4			
4.1	This manuscript entitled "Prevalence and predictors of medication non-adherence among people living with multimorbidity: a systemic review and meta-analysis" added to the literature in examining the medication non-adherence among individuals with multimorbidity, regardless of medications and conditions to be investigated. The strengths of this work include identifying an inclusive list of eligible studies. While I appreciate the efforts that investigators devoted to this work, quite a few major issues substantially dampen my enthusiasm for this study.	Many thanks to this reviewer for your helpful comments, which are addressed below and in the revised manuscript.	n/a
4.2	As investigators acknowledged in the study, medication non-adherence varied significantly by medications and conditions to be investigated. Furthermore, medication non-adherence can be impacted by many other factors like the reimbursement policies distinct in countries/regions. Ignoring the substantial heterogeneity of the included studies and providing a pooled overall non-adherence prevalence is not informative from my perspective. Investigating a more granular level of medication non-adherence based on the patients, conditions, and medications is more of interest in my point.	As also reflected upon further in 4.8 below, the review did not aim to investigate adherence at a granular level by looking at specific conditions and treatments, as this approach is considered to be at odds with the positioning of multimorbidity as a common experience among a diverse group of individuals. Such an approach risks a return to the single-disease focus that has dominated the literature.[8] The aim of our quantitative synthesis is to maintain a general and inclusive approach to multimorbidity, albeit among a diverse group of participants, in order to represent the complexity that is inherent in multimorbidity research.	n/a

		Please also see point 4.8 below in relation to further sub-group analyses e.g. stratifying by reimbursement policy, demographics etc.	
4.3	There are several notable major concerns in the manuscript needed to be clarified in order to ensure the validity of the findings. Areas could be improved to strengthen the manuscript and study are listed below.	We have addressed these below and in the revised manuscript where applicable.	n/a
4.4	1. (Page 3, abstract, methods): Investigators stated the databases that were searched for relevant studies. Why did not the investigators search the Cochrane Library (https://www.cochranelibrary.com/cdsr/about-cdsr) ? It is an important data source to check whether any similar systemic review and meta-analysis has been conducted previously. If so, investigators should mention that in the introduction to clarify what incremental values that this manuscript will provide. There may be some systemic review studies that have investigated the similar research question, but I did not see investigators mentioning them in the introduction. Failing to do that weakens the importance of this study. Furthermore, the investigators may identify extra eligible studies through reviewing the Cochrane reviews.	Prior to finalising our systematic review aims, we searched in PROSPERO[21], the prospective register of systematic reviews, to determine if a similar review had been conducted previously or was currently being conducted. No similar review was found. We have updated the manuscript to explain this. Similarly, to our knowledge, no Cochrane review had been published reporting prevalence of non-adherence among people with multimorbidity; therefore the Cochrane Library database was not formally searched as part of this review. Our choice of databases to search was determined by existing published reviews researching related topics, i.e. prevalence and predictors of medication non-adherence, e.g. [22].	Prior to finalising our systematic review aims, we searched in PROSPERO[21], the prospective register of systematic reviews, to determine if an equivalent review had been conducted or was currently being conducted. No equivalent review was found.

4.5	2. (Page 3, abstract, methods): The time span of this review was between 2009 and April 2019 as indicated in the abstract. I understand that the purpose was to align with the protocol published in 2019. However, given the date of reviewing this study (April 2021), I strongly suggest investigators performing an updated search looking till the current date	As also outlined to the editor, we acknowledge that this suggestion to update the search may strengthen the relevance of the study in light of the months passed between our original submission and revision. However, considering the significant time and human resources required to screen and integrate newer studies into a revision, this task is not considered feasible within the current project which forms one part of a PhD programme. We recognise that BMJ Open has previously accepted systematic reviews with a similar time period between the search date and submission date, e.g.: Zaragoza-Martí A, Cabañero-Martínez M, Hurtado-Sánchez J*, et al* Evaluation of Mediterranean diet adherence scores: a systematic review *BMJ Open* 2018;**8:**e019033. doi: 10.1136/bmjopen-2017-019033 [search completed Dec 15, manuscript submitted Aug 17] Amiri Farahani L, Asadi-Lari M, Mohammadi E*, et al* Community-based physical activity interventions among women: a systematic review *BMJ Open* 2015;**5:**e007210. doi: 10.1136/bmjopen-2014-007210 [search completed Mar 13, manuscript submitted Nov 14]	n/a
4.6	3. (Page 5, introduction): Investigators did not mention any of previous systemic review or meta-analysis studies that examined the medication non-adherence among people with	To our knowledge, no existing systematic review has synthesised studies reporting prevalence and/or predictors of medication non-adherence among people living with multimorbidity.	To our knowledge, a synthesis of the prevalence and predictors of medication non-adherence among people with multimorbidity has not been previously reported.

	multimorbidity. It is unclear what extra values that this study will provide if there is such type of study. Suggest investigators adding some contents to clarify that. Thanks!	We have updated the introduction section to highlight this gap in the literature that is addressed by our review.	
4.7	4. (Page 6-7, data extraction and quality assessment): Investigators discussed that they would assess the quality of each study, but no studies would be excluded based on the quality. An overall view of study quality was presented in Figure 2. However, I am not sure of the value of assessing the study quality if the investigators did not exclude those studies with low-quality or did not do any subgroup analysis in reporting the results in high- and low-quality studies separately. From my perspective, reporting and synthesizing results from low-quality studies is pointless due to their untrustworthy findings. I understand that investigators would like to include an inclusive list, but a balance between completeness and study quality needs to be achieved. If investigators want to include all the studies, I suggest stratifying the studies based on quality and then reporting the results separately. In addition, Figure 2 just provided an overall view of study quality. Suggest adding one more column in the study characteristics table in the supplement to indicate the quality of each study. Last, it is difficult to read the assessment items in Figure 2. Please make the font size bigger.	The quality assessment tool we used in this review takes a checklist approach to avoid assumptions about the relative importance of individual items[23]. We elected not to attribute a single summary score to each study, as the measure used has not been validated to produce a single score. Use of single summary scores for quality assessment in systematic reviews has been judged as problematic in the existing literature, given that individual items may be statistically independent.[24] This therefore precludes exclusion/stratification of studies based on quality and incorporation of quality assessments into the analysis [23]. However, to enhance the value of the quality assessment in this review, we have added a section to the manuscript outlining future recommendations based on the findings of the quality assessment, as recommended as a means to incorporate findings from quality assessment into systematic reviews.[23] The size and style of the font in figure 2 has been changed to improve the clarity of this figure.	The quality assessment produced a number of recommendations for future research exploring non-adherence among people with multimorbidity. Of note, while almost all studies (94%) were judged to have used non-adherence measures that were reproducible, 16% of studies used methods that were not considered to be valid. The validity of non-adherence assessment should be a priority for future related investigations. Furthermore, almost 30% of studies were judged not to have reduced serious recall bias. It has been suggested that the appropriate period for retrospective self-report of non-adherence should be balanced between optimising recall and capturing instances of infrequent non-adherence.[25] The optimum recall period for measures of multiple medication non-adherence should be determined.
4.8	5. (Page 7, data analysis): It claimed that “Meta-analysis was conducted with studies which recruited participants as ‘people living with >= 2 chronic conditions’ and not studies which recruited	As mentioned in section 4.2 above, and 5.2 below, we acknowledge that the meta-analysis is impacted by the variation in adherence found across conditions and	From eight studies considered suitable for meta-analysis, 43% (95% CI 34%-51%) of participants with multimorbidity were non-adherent to medications. Non-adherence ranged from 16.4%[28] to 61.4%,[29] with

	participants with combinations of specific conditions". I agree that a quantitative synthesis including all studies was not appropriate given the substantial heterogeneity between studies. However, I am not sure why meta-analysis was conducted only for these studies as well as the value of results. The investigators claimed that these studies reflected the most inclusive definition of multimorbidity, but based on my observations, the medication non-adherence varies substantially by medications and conditions to be investigated. Many other factors will also affect the medication non-adherence (e.g., reimbursement policy in the corresponding country/region, patient demographics). Providing an overall pooled prevalence of medication non-adherence regardless of specific medications and indications is not informative from my perspective. I cannot see the value of these findings. Investigators could look into a more granular level of medication non-adherence in different types of patients, conditions, and medications. I think those results are more of interest compared to the present results in the manuscript. Suggest investigators conducting more subgroup analyses.	medications, as well as methodological features of the included studies, including sample size and method of adherence measurement. To emphasise the range of non-adherence found across studies included in the meta-analysis, we have added more detail to the discussion section of the revised manuscript as a caution for readers interpreting the meta-analysis results. The heterogeneity of multimorbidity itself presents inherent challenges for quantitative synthesis, given the diversity of participants not only between, but also within individual studies.[26] Our meta-analysis aims to attempt this type of synthesis, while attaching caution around interpretation, as "*to abort the meta-analysis because of heterogeneity fails to address the question of how to synthesise the data*".[27] In relation to investigating adherence at a more granular level by looking at specific conditions and treatments, this approach is at odds with the positioning of multimorbidity as a common experience among a diverse group of individuals, and risks regression towards the single-disease focus that has dominated until recently.[8] The aim of our synthesis, therefore, is to maintain a general approach to multimorbidity, albeit among a diverse group, in order to represent the complexity that is inherent in multimorbidity research.	the lowest estimate arising from a single-item self-report measure,[28] and the highest derived from pharmacy data.[29] One study in the meta-analysis contributed to 76% of the sample size.[30] These characteristics of individual studies must be considered when interpreting findings of the quantitative synthesis. ***Addition to data sharing statement:*** Our systematic review produced a large amount of information and the arising database is available for future collaboration on additional analyses. Please contact the corresponding author with any inquiries.

		Nevertheless, we recognise the potential to complete a wide variety of very interesting subgroup analyses using the wealth of data arising from this systematic review, including using demographic data and reimbursement policy data to stratify analyses; however this is not considered possible within the constraints of a single journal article. For this reason, we will make our comprehensive database available to the research community for collaboration on future sub-analyses, as has previously been done in a Cochrane review of medication adherence interventions reporting a similarly large number of studies influenced by a wide range of factors.[9]	
4.9	6. (Page 8, study characteristics; Page 9, prevalence of medication non-adherence): It stated in "Study characteristics" that 19 studies recruited participants using an inclusive definition of multimorbidity, but under "Prevalence of medication non-adherence" only 8 studies were included in the meta-analysis. Could investigators clarify why 11 studies were excluded from the meta-analysis?	The meta-analysis only included studies which reported prevalence of non-adherence as a proportion of the total study sample. The remaining 11 studies were not included in the meta-analysis, as the *average* rate of non-adherence across participants was reported in these studies and not *prevalence* of non-adherence.	n/a
Reviewer 5			
5.1	Overall, a very thorough review and well written manuscript. The inclusion of search criteria and individual study data (even R code) demonstrate both meticulousness and transparency on behalf of the authors. My minor comments are the following:	Many thanks to this reviewer for your helpful comments, which are addressed below and in the revised manuscript.	n/a

5.2	Figure 3 – results from the random effects model seem to just point to the largest study dominated by 10x the population of the other studies, but the non-adherence range among all 8 studies is considerable (<20% to 60%). Please walk the readers through the interpretation of this result give such high variability for a small number of studies.	The discussion of the results of the meta-analysis has been revised to highlight the range of non-adherence and potential source of this difference between studies. The contribution of one study to a substantial proportion of the overall sample size has also been emphasised to readers in this section of the manuscript.	From eight studies considered suitable for meta-analysis, 43% (95% CI 34%-51%) of participants with multimorbidity were non-adherent to medications. The range of non-adherence across these eight studies was from 16.4%[28] to 61.4%,[29] with the lowest estimate arising from a single-item self-report measure,[28] and the highest derived from pharmacy data.[29] Of note, one study in the meta-analysis contributed to 76% of the sample size in the meta-analysis.[30] These characteristics of individual studies must be considered when interpreting findings of the quantitative synthesis.
5.3	In the section, "Medication non-adherence over time" please summarize the endpoints for non-adherence across studies (e.g., what fraction of studies measured adherence in a few months, 6 months, 1 year, etc.), as it's difficult to interpret increases and decreases without knowing the denominator	Thanks to the reviewer for this suggestion which we agree aids the interpretation of changes in adherence over time. We have updated the manuscript to report the length of follow-up for each study reported in this section.	Among studies reporting prevalence of non-adherence over time, respective lengths of follow-up were three months,[31] six months,[32] 48 weeks,[33] one year,[34,35] 16 months,[36] two years,[37] 3 years[38] and 5 years.[39]
5.4	The discussion section raises the point that medication frequency or number of tablets of course impacts medication adherence. It would be helpful to summarize in the results any findings regarding differences in adherence among conditions that require higher medication frequency or tablet count, compared to conditions with lower medication frequency/table counts.	This is a most interesting point. As we did not have access to individual patient data regarding individual treatment regimens, it is beyond the scope of this review to perform these analyses. The manuscript has been updated to reflect this.	Indeed, it has been reported that particular combinations of conditions are associated with higher dosing frequencies in multimorbidity, with potential implications for treatment burden,[40] and therefore possibly influencing medication adherence. Mode of drug delivery may also be an influencing factor, with different modes of delivery varying in convenience and, therefore, adherence.[40]Due to lack of individual patient data regarding treatment regimens, it is beyond the scope of this review to explore this further.

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VERSION 2 – REVIEW

REVIEWER	Fowler Davis, Sally Sheffield Hallam University, Faculty of health and Wellbeing
REVIEW RETURNED	13-Jul-2021

GENERAL COMMENTS	Thanks for the opportunity to review this interesting review that addresses an important topic related to medication adherence. The focus on multiple morbidity is, to my knowledge, novel and a relevant focus, although I was curious that the results re-focussed on separate conditions. The literature selected is extensive and well managed with limitations reported as a minor risk to the efficacy of the findings. The authors may consider a couple of critical issues related to the discussion. The first relates to the management of non-adherence in practice and the limited range of measures available to optimise medications. For example the de-prescribing processes are not consistent or sufficiently linked to poly pharmacy adherence and this reflection on the evidence would deepen the relevance to clinical pharmacy. Similarly the tentative focus on the age demographic and the lack of evidence for worse outcomes was of interest but this needed qualifying given the limited range of data related to population segmentation. I have limited comment on the methods which I found robust and well justified.

REVIEWER	Calderón-Larrañaga, Amaia Karolinska Institutet
REVIEW RETURNED	21-Jul-2021

GENERAL COMMENTS	The authors have adequately addressed my comments.

REVIEWER	Zhou, Lili AbbVie Inc, Health economics and outcome research

REVIEW RETURNED	10-Jul-2021

GENERAL COMMENTS	The authors have well addressed my previous comments. No further comment for the revised manuscript!

REVIEWER	Belli, Hayley NYU Langone Health, Population Health
REVIEW RETURNED	07-Jul-2021

GENERAL COMMENTS	The authors appropriately addressed all of my comments and concerns.

VERSION 2 – AUTHOR RESPONSE

Response to reviewers' comments:

Reviewer: 5
Dr. Hayley Belli, NYU Langone Health
Comments to the Author:
The authors appropriately addressed all of my comments and concerns.
Many thanks for reviewing our revised manuscript.

Reviewer: 4
Dr. Lili Zhou, AbbVie Inc
Comments to the Author:
The authors have well addressed my previous comments. No further comment for the revised manuscript!

Thank you very much for reviewing our revised manuscript.

Reviewer: 1

Dr. Sally Fowler Davis, Sheffield Hallam University, Sheffield Teaching Hospitals NHS Foundation Trust
Comments to the Author:

Thanks for the opportunity to review this interesting review that addresses an important topic related to medication adherence. The focus on multiple morbidity is, to my knowledge, novel and a relevant focus, although I was curious that the results re-focussed on separate conditions. The literature selected is extensive and well managed with limitations reported as a minor risk to the efficacy of the findings.

Many thanks to the reviewer for this positive feedback.

The authors may consider a couple of critical issues related to the discussion. The first relates to the management of non-adherence in practice and the limited range of measures available to optimise medications. For example the de-prescribing processes are not consistent or sufficiently linked to poly pharmacy adherence and this reflection on the evidence would deepen the relevance to clinical pharmacy.

Many thanks for this suggestion. We have added a point to our discussion to highlight that while deprescribing may be necessary, it is not always sufficient to promote adherence to multiple medications, potentially producing a clinical dilemma for clinicians. Please see page 11 of revised manuscript.

A recent systematic review concluded that deprescribing interventions do not consistently improve adherence,[4] suggesting an equivocal relationship between medication burden and non-adherence. This highlights the clinical dilemma faced by clinicians where deprescribing may be a necessary but not sufficient therapeutic response.

Similarly the tentative focus on the age demographic and the lack of evidence for worse outcomes was of interest but this needed qualifying given the limited range of data related to population segmentation.

Thanks for this feedback. We recognise the limited range of data on age from our synthesis of prevalence and have acknowledged this in our discussion. This section of the discussion has been reworked to highlight the relevance of our synthesis of predictors of non-adherence, with no studies reporting age as a significant predictor. References to studies from our synthesis of predictors have been more clearly signposted in this section to highlight links between our own findings and existing literature. Please see page 12 of revised manuscript.

We did not stratify our analysis of prevalence by age, potentially limiting interpretation of our findings. Notably, age was not found to be a significant correlate in our synthesis of predictors, and it has been suggested previously that there is no consensus on age as a predictor of non-adherence.[5] Nevertheless, it may be plausible that variables are differentially associated with non-adherence at different life stages. For example, among reviewed studies, Domino and colleagues[6] reported lower levels of non-adherence in people with more conditions, with a mean age among participants of 43 years; Schüz and colleagues found higher non-adherence among participants with more conditions, with a mean age of 73 years.[7] Exploring how non-adherence is variably predicted by clinical and psychosocial influences across the lifespan could inform intervention development which is appropriately targeted according to life stage.

I have limited comment on the methods which I found robust and well justified.

Many thanks for this positive feedback.

Reviewer: 2
Dr. Amaia Calderón-Larrañaga, Karolinska Institutet

Comments to the Author:
The authors have adequately addressed my comments.

Many thanks for reviewing our revised manuscript.

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